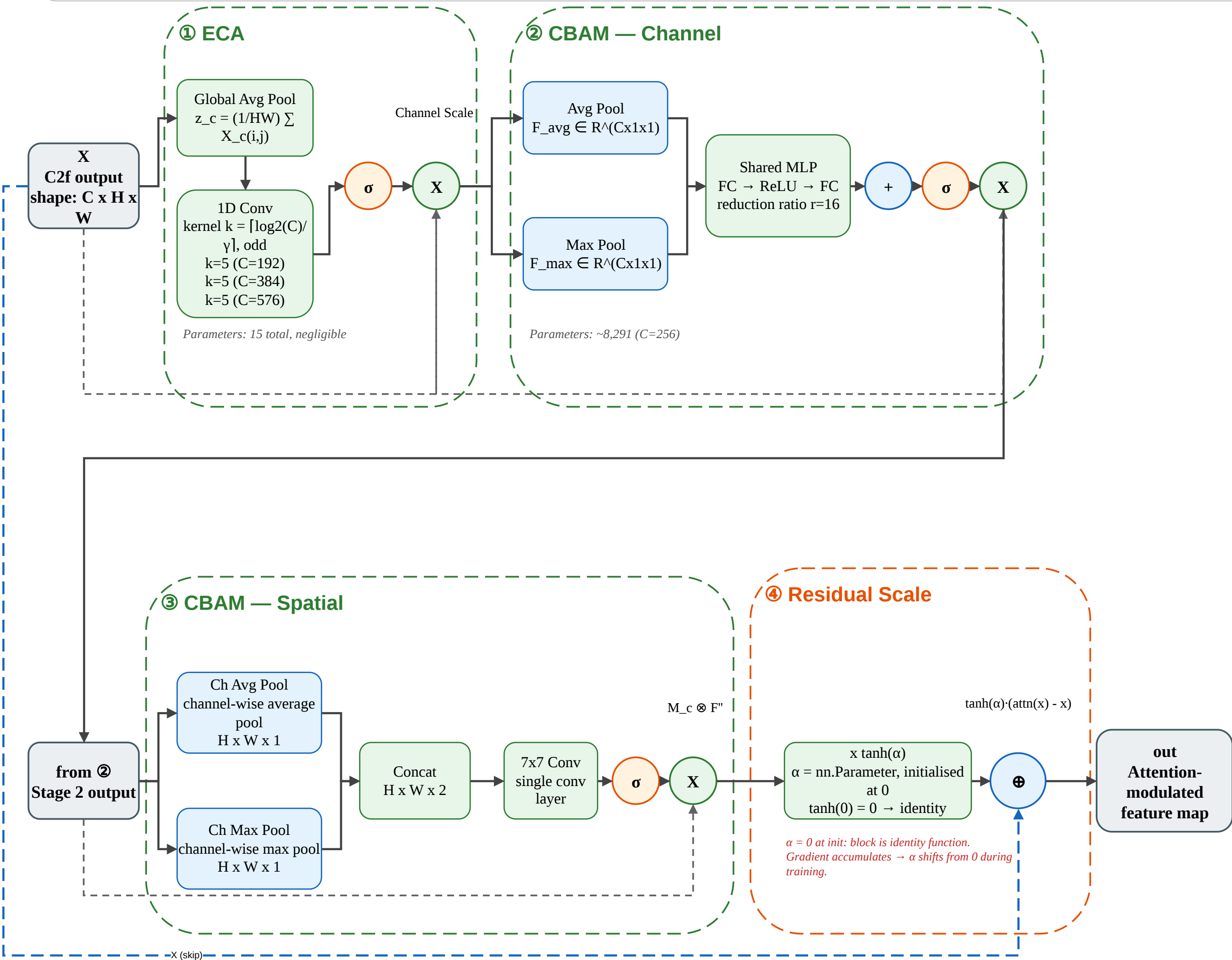


(1) $z_c = (1/HW) \sum_{i,j} X_c(i,j)$
[ECA pooling]

(2) $M_c = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F)))$
[CBAM channel]

(3) $\text{out} = x + \tanh(\alpha) \cdot (\text{CBAM}(\text{ECA}(x)) - x)$
[Full block]



Total ECA parameters: 15

Total CBAM parameters: 8,291 per
injection point

Total ECA+CBAM overhead: +64,818
params, 0.25% of YOLOv8m